ITK-SNAP as an Agent-Callable Tool: Expert Human Correction as a Resumable, Audited Pipeline Step

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Body word count: **490 / 500** *(title, section headings and keywords are excluded per submission rules §4)*

Deadline: **July 24, 2026, 11:59 PM PST** · Portal EventKey: QRFBVSUS · *Not blind: institution / tool names allowed.*

Keywords

human-in-the-loop; interactive segmentation; agentic AI; Model Context Protocol; provenance; foundation models; TotalSegmentator; open-source toolkits

Problem Statement

87 words

Automatic segmentation models are increasingly accurate but still imperfect, so clinical and research pipelines still rely on expert review and correction. Yet that judgment remains confined to interactive software: a person opens an image, paints a fix, and saves a file, while the reasoning and the exact change are lost. An automated pipeline has no direct way to *call* on an expert as a defined step and receive a result it can read back. Human review is essential in practice yet largely absent from our software interfaces.

Approach / What You Built

166 words

We made ITK-SNAP — a 20-year-old open-source segmentation application with over a million downloads — callable by an AI agent through the Model Context Protocol (MCP), a standard way to connect programs to external tools. The agent runs a segmentation model (TotalSegmentator, hosted on our open itksnap-dls server) and writes its proposal into a saved ITK-SNAP workspace in the background. When a case needs review, the agent reopens that workspace in an interactive session for a person to correct. Every edit, by agent or person, returns a structured record of what changed, not just a saved file (Table 1). Two decisions make this practical. First, the record is captured once, in one format — whether ITK-SNAP is edited in the background or interactively — so none of its eleven editing tools had to change. Second, because all state lives in the workspace file, the step can be paused and resumed: a pipeline can prepare a case before anyone is involved, and the record persists across sessions.

Demo or Evidence of Function

96 words

The code (two open-source repositories) and a short video walkthrough are linked with the submission. We verified the full sequence on a GPU: the agent ran TotalSegmentator on a body CT (48 anatomically correct thoracic structures), wrote one structure into an ITK-SNAP workspace, and read back the record (Table 1). A person opened that workspace in ITK-SNAP and corrected the proposal with the paintbrush; the agent received an identical record automatically labeled as human-made (`actor: human`) — because the agent reads that label, any edit it did not make is attributed to the person (Figure 1).

Clinical or Operational Impact

47 words

Expert review becomes a pipeline step that can be scheduled, paused, and resumed. In a large automated study, a person reviews only uncertain cases, and each correction is recorded — what changed and who made it — in a form reusable to retrain models and audit quality.

Current Stage

49 words

Prototype, open-source, not a product. The propose–apply–record workflow works from start to finish; the correction in the walkthrough was made by a developer, not a clinician, so it tests the mechanism rather than clinical judgment. Routing only low-confidence cases to a person, and a packaged release, are in progress.

What Feedback You're Seeking

45 words

Which fields in the change record matter most for retraining and record-keeping? What confidence measures would you trust to decide between accepting a result and routing it to a person? And where would a callable expert-correction step fit into your existing annotation and quality-control workflows?

Figures

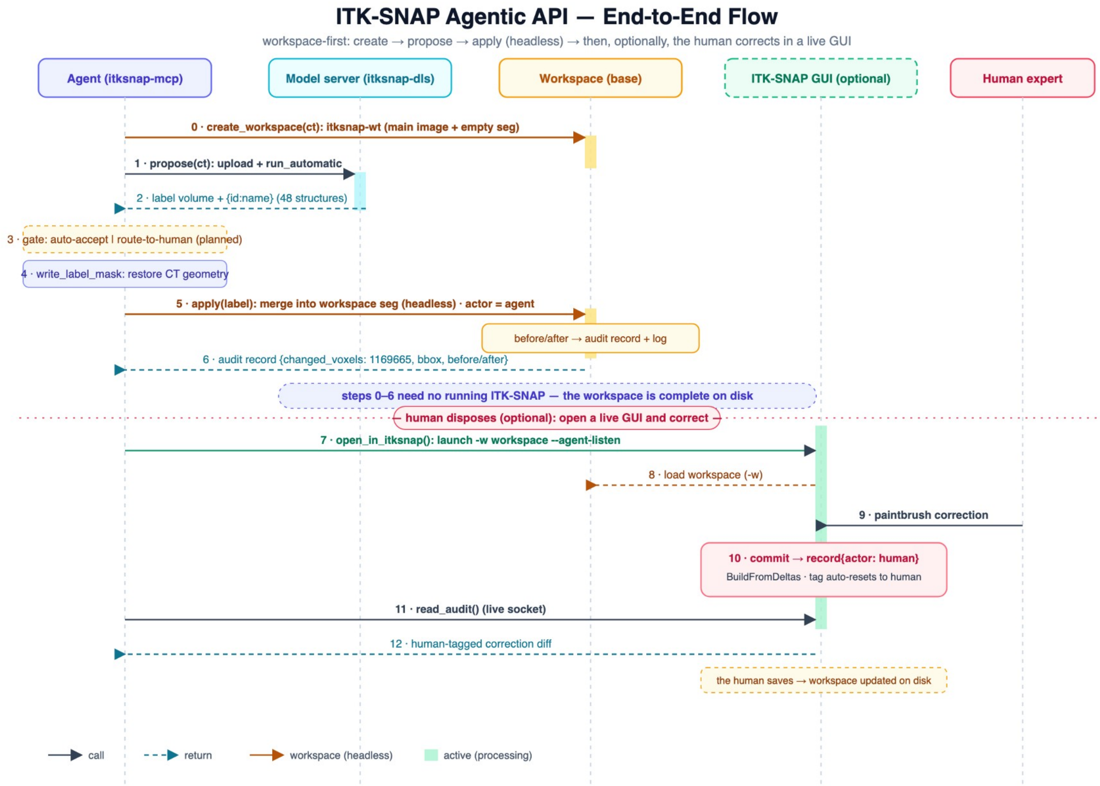

Figure 1. One complete run, centered on the workspace file. The agent creates an ITK-SNAP workspace, gets an automatic proposal from the model server, and writes it into the workspace in the background, with no ITK-SNAP window open (steps 0–6). When a case needs review it opens the workspace in ITK-SNAP for a paintbrush correction; every saved edit returns the same structured record, labeled agent or human.

What comes back: the audit record for one committed edit

values are from the verified end-to-end run — a proposed left upper lung lobe applied into an ITK-SNAP workspace

Field	Live run — agent apply	Why a downstream consumer needs it
`op`	`"Agent apply (proposal)"`	which editing tool produced the change
`timestamp`	`"2026-07-19T02:26:20Z"`	ISO-8601 UTC — orders edits within a session
`actor`	**`"agent"`**	**machine output or expert judgment — decides whether these voxels are training-grade ground truth**
`changed_voxels`	`1169665`	magnitude of the correction — a cheap proxy for how far off the model was on this case
`rle_count`	`— (see note)`	edit complexity: one contiguous fill vs. many scattered touch-ups
`time_point`	`0`	which frame of a 4D series the edit applies to
`bbox`	`min [84, 2, 0]` `max [247, 189, 180]`	where the correction is — a crop region for focused review or targeted re-training
`before_counts`	`{"0": 1169665}`	what was overwritten — background vs. another label, which reveals the model's confusion pairs
`after_counts`	`{"1": 1169665}`	what those voxels became

Captured at a single commit chokepoint, so the record is identical whichever tool made the edit — paintbrush, polygon, 3D spray, threshold, or an agent's apply.
A human correction returns the same structure with **`actor: "human"`**; nothing else about the record changes.
Note: every field listed is emitted with every record; the `rle_count` value was not captured in the logged excerpt of this particular run.

Table 1. What comes back: the record for one saved edit. Values from the verified run (a proposed left upper lung lobe). The same fields are returned whether the edit is made in the background or corrected interactively in ITK-SNAP; a human correction returns an identical record labeled actor: human.

Submission links

Not part of the 500-word body — paste into the portal's demo / repo / video fields.

Agent code (Python): `https://github.com/jilei-hao/itksnap-mcp`

Audit engine, ITK-SNAP fork (C++): `https://github.com/jilei-hao/itksnap`

Demo video (YouTube): `[add URL after upload]`

Demo video — YouTube title and description

Metadata for the video upload, not part of the abstract.

Title — primary (68 characters, displays in full):

```
Auditable Human-in-the-Loop Segmentation with ITK-SNAP and AI Agents
```

Title — alternative (mirrors the abstract title):

```
ITK-SNAP as an Agent-Callable Tool: Auditable Human Correction in an
AI Pipeline
```

Description (paste verbatim into YouTube):

```
An open-source prototype that lets an AI agent call ITK-SNAP as a tool
and treats expert human correction as an auditable, resumable step in
an automated segmentation pipeline.

In this walkthrough, the agent runs an automatic segmentation model
(TotalSegmentator) on a body CT and writes the result into a saved
ITK-SNAP workspace in the background — with no window open. A person
then opens the same workspace in ITK-SNAP and corrects the result with
the paintbrush. Every edit, by agent or human, returns a structured
```

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record of what changed and who changed it, so corrections become
reusable, attributable data.

Code:
• Agent (Python): https://github.com/jilei-hao/itksnap-mcp
• ITK-SNAP fork (C++): https://github.com/jilei-hao/itksnap

Presented at the SIIM-CAIMI26 AI Builder Showcase.
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